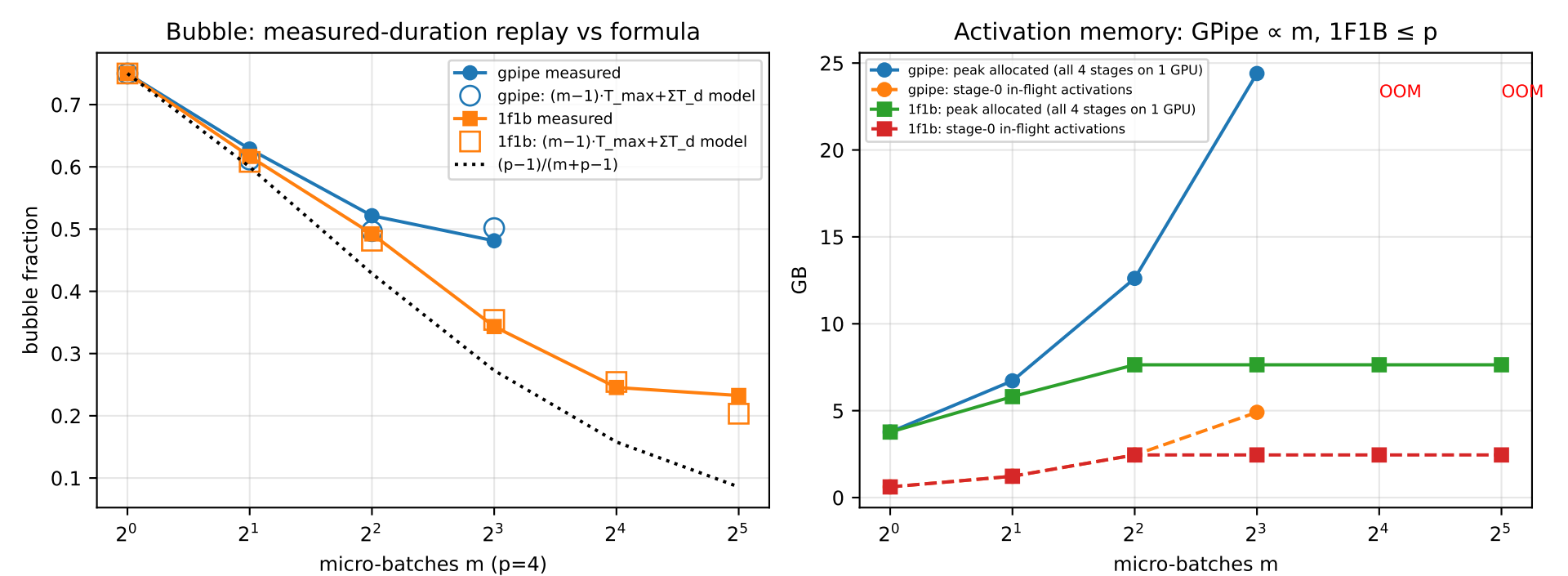
Bubble: measured-duration replay vs formula
gpipe measured
gpipe: (m−1)·T_max+ΣT_d model
1f1b measured
1f1b: (m−1)·T_max+ΣT_d model
(p−1)/(m+p−1)
bubble fraction
micro-batches m (p=4)
Activation memory: GPipe ∝ m, 1F1B ≤ p
gpipe: peak allocated (all 4 stages on 1 GPU)
gpipe: stage-0 in-flight activations
1f1b: peak allocated (all 4 stages on 1 GPU)
1f1b: stage-0 in-flight activations
OOM
OOM
GB
micro-batches m